

## Cover letter to the Editors of *Quantum*

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Dear Editors,

On April 23, 2026, we submitted our manuscript “**Blind Catalytic Quantum Error Correction: Target-State Estimation and Fidelity Recovery Without A Priori Knowledge**”, authored by Hikaru Wakaura and Taiki Tanimae, for consideration as an Article in *Quantum*. The submission consists of the main manuscript (~25 pages, 18 figures, 9 tables, single bibliography) and an open-source Python package release (`blind_cqec`, MIT license, available at [https://github.com/deeptell-inc/blind\\_cqec\\_pkg](https://github.com/deeptell-inc/blind_cqec_pkg)) that ships a 61-test reproducibility suite locking in every numerical value reported in the paper.

**What we discovered.** Catalytic quantum error correction (CQEC) is a recent threshold-free recovery framework that has so far required a strong assumption: the ideal target state must be known exactly to construct the catalytic map. We discovered that this assumption can be removed almost entirely. By prepending a classical pre-estimator to the catalytic recovery, we obtain a two-stage protocol—blind CQEC—whose recovery fidelity satisfies a clean analytical bound,  $F_{\text{rec}} \geq 1 - 2 \|\hat{\rho}_{\text{est}} - \rho_{\text{target}}\|_1$ , with an empirical Lipschitz constant of  $0.98 \pm 0.02$ . The bound is matched by a near-perfect linear correlation between estimation and recovery fidelities ( $r > 0.99$  across 84 conditions), implying that the entire performance of blind CQEC is governed by a *single classical estimation problem*. A natural consequence is an analytical crossover dimension  $d^* \approx 25\text{--}40$  separating a noise-model-free regime (coherence maximization) from a noise-informed regime (channel inversion), with a tunable hybrid that bridges the two.

**Why this matters.** The result reframes blind error correction as a classical optimization problem, opening a tractable design space that prior coherence-resource-theoretic protocols left implicit. It also fills a gap in the existing error-handling stack: standard QEC requires pre-encoding and a noise threshold; quantum error mitigation (ZNE, PEC, virtual distillation) returns corrected expectation values rather than the state itself and pays exponential overhead in circuit depth or qubit count. Blind CQEC sits between these two—post-hoc, threshold-free, single-copy, and state-level—which is precisely the regime relevant to variational and iterative quantum algorithms whose output is unknown to the correction module. A noisy VQE simulation for  $\text{H}_2$  already shows a  $3.4\times$  energy-error reduction without any knowledge of the ansatz state, and a circuit-level sanity check on `qiskit-aer` confirms that the density-matrix predictions transfer to small noisy circuits.

**Why *Quantum*.** *Quantum*’s combined emphasis on theoretical rigor, comprehensive numerical evidence, and open scientific practice fits this work directly. The manuscript provides a proven Lipschitz bound, a closed-form crossover dimension, and ensemble benchmarks on Haar-random states up to  $d = 256$ . The accompanying `blind_cqec` package is released under the MIT license with full continuous-integration coverage, deterministic seeds, and a numerical regression test suite, supporting the reproducibility standards *Quantum* explicitly champions. The companion preprints on which we build are all publicly available (arXiv:2603.25774, doi:10.21203/rs.3.rs-9366084, arXiv:2603.11568), so the manuscript is fully self-contained for review. We believe the breadth of evidence, the analytical content, and the open-science deliverables make this submission well-aligned with *Quantum*’s scope and standards, and that it will be of broad interest

to the quantum-error-correction, quantum-error-mitigation, and resource-theory communities.

**Suggested referees.**

1. **Ryuji Takagi**, Nanyang Technological University, Singapore ([ryuji.takagi@ntu.edu.sg](mailto:ryuji.takagi@ntu.edu.sg)). Co-author of the foundational PRL on catalytic coherence transformations; primary expert on the resource-theoretic backbone.
2. **Mark M. Wilde**, Cornell University, USA ([wilde@cornell.edu](mailto:wilde@cornell.edu)). Quantum Shannon theory, channel fidelity bounds, asymmetry measures.
3. **Andreas Winter**, Universitat Autònoma de Barcelona, Spain ([andreas.winter@uab.cat](mailto:andreas.winter@uab.cat)). Catalytic resource theories, asymptotic transformations.
4. **Christophe Piveteau**, IBM Research Zürich, Switzerland ([cpv@zurich.ibm.com](mailto:cpv@zurich.ibm.com)). Probabilistic error cancellation; well-placed to evaluate the QEM comparison.
5. **Bálint Koczor**, University of Oxford, UK ([balint.koczor@maths.ox.ac.uk](mailto:balint.koczor@maths.ox.ac.uk)). Virtual distillation, error mitigation, near-term VQE.
6. **Daniel Stilck França**, ENS Lyon / CNRS, France ([daniel.stilck-franca@ens-lyon.fr](mailto:daniel.stilck-franca@ens-lyon.fr)). Quantum state estimation, noise-aware tomography, copy complexity.
7. **Tim Byrnes**, NYU Shanghai, China ([tim.byrnes@nyu.edu](mailto:tim.byrnes@nyu.edu)). Co-author of the Purification QEC framework cited in our comparison.
8. **Martí Perarnau-Llobet**, University of Geneva, Switzerland ([marti.perarnaullobet@unige.ch](mailto:marti.perarnaullobet@unige.ch)). Catalysis and asymmetry resource theory; quantum thermodynamics.
9. **Iman Marvian**, Duke University, USA ([iman.marvian@duke.edu](mailto:iman.marvian@duke.edu)). Symmetry/asymmetry resource theory, covariant operations.
10. **Patrick Hayden**, Stanford University, USA ([phayden@stanford.edu](mailto:phayden@stanford.edu)). Quantum information theory, fidelity-based recovery and decoupling.

**Non-preferred referees.** None.

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**Statements.** This work has not been published or submitted elsewhere. Both authors have read and approved the submission. We declare no competing interests.

We thank you for your consideration and look forward to the referees’ feedback.

Sincerely,

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QIRI Inc., Tokyo